

GRADE 11 • ALGEBRA II • TOPIC 6

Lesson 6-4

Logarithmic Functions

Exponential and Logarithmic Functions

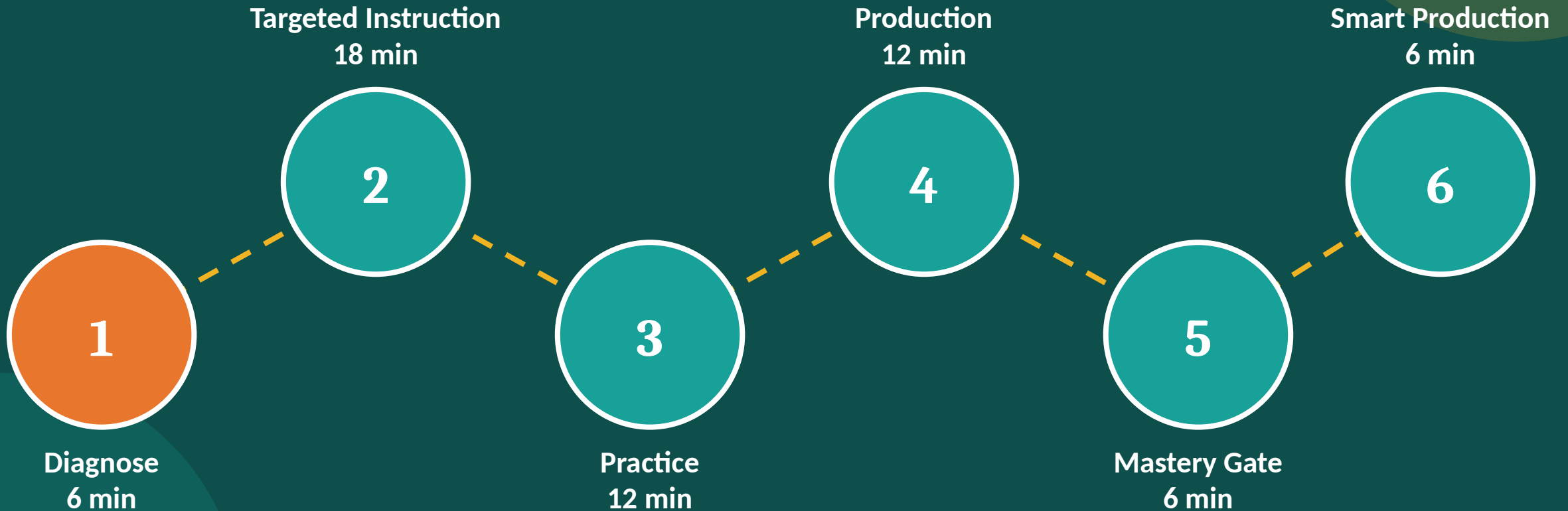
Mr Malek Thiab

$$y = \log_3 x, \quad y = 3^x$$

Journey of 6 stations, 60 minutes

Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Identify key features of logarithmic functions.
2. Use inverse properties to analyze logarithmic and exponential functions.
3. Graph logarithmic functions and interpret their key features.
4. Write and interpret the inverses of exponential and logarithmic functions.

VOCABULARY

inverse relationship

logarithmic function

ESSENTIAL QUESTION

How is the relationship between logarithmic and exponential functions revealed in the features of their graphs?

MATHEMATICAL PRACTICES MP.4 · MP.7

Standards

HSF.IF.B.5 Relate the domain of a function to its graph and, where applicable, to the quantitative relationship it describes.

HSF.IF.B.6 Calculate and interpret the average rate of change of a function (presented symbolically or as a table) over a specified interval. Estimate the rate of change from a graph.

HSF.IF.C.7.E Graph exponential and logarithmic functions, showing intercepts and end behavior, and trigonometric functions, showing period, midline, and amplitude.

HSF.IF.C.9 Compare properties of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).

HSF.BF.B.3 Identify the effect on the graph of replacing $f(x)$ by $f(x) + k$, $k f(x)$, $f(kx)$, and $f(x + k)$ for specific values of k (both positive and negative); find the value of k given the graphs. Experiment with cases and illustrate an explanation of the effects on the graph using technology. Include recognizing even and odd functions from their graphs and algebraic expressions for them.

HSF.BF.B.4 Find inverse functions.

(+)HSF.BF.B.4.C Read values of an inverse function from a graph or a table, given that the function has an inverse.

STATION 1 • DIAGNOSE • 6 MIN

Start here: three quick checks

1

Solve $3^x = \frac{1}{9}$

2

If $f(x) = 2x + 1$, find $f^{-1}(x)$

3

Average rate of change of x^2 on $[1, 3]$

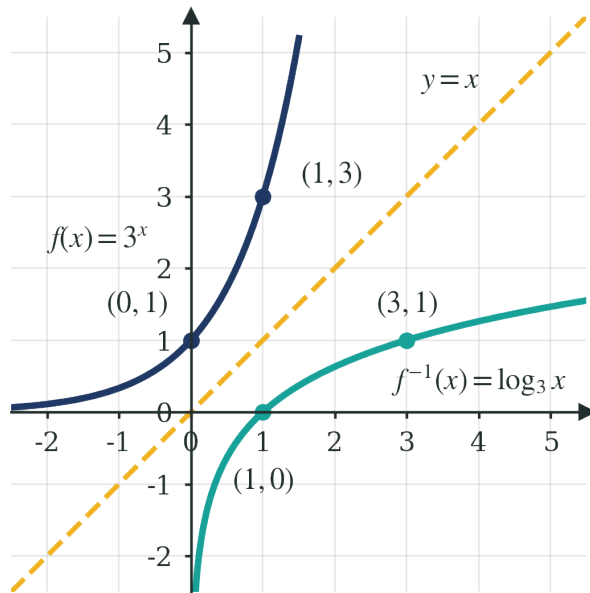
Done when

All three answers are written without a calculator, and you can say in one sentence what swapping inputs and outputs does to a function.

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Exponentials and logarithms are inverses

$$f(x) = 3^x \iff f^{-1}(x) = \log_3 x \quad (a, b) \text{ on } f \iff (b, a) \text{ on } f^{-1}$$



EXAMPLE A

Table of $f(x) = 3^x$: read $f^{-1}(3)$
and $f^{-1}(\frac{1}{3})$

$$f(1) = 3 \Rightarrow f^{-1}(3) = 1$$

$$f(-1) = \frac{1}{3} \Rightarrow f^{-1}(\frac{1}{3}) = -1$$

EXAMPLE B

Find the inverse of $f(x) = 3^x + 2$

$$y = 3^x + 2 \rightarrow x = 3^y + 2$$

$$3^y = x - 2$$

$$f^{-1}(x) = \log_3(x - 2)$$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Key features of the logarithmic function

Domain: $x > 0$, Range: all real y Asymptote $x = 0$, x -intercept $(1, 0)$

EXAMPLE A

Evaluate $\log_3 \frac{1}{9}$ and $\log_3 27$

$$3^{-2} = \frac{1}{9} \Rightarrow \log_3 \frac{1}{9} = -2$$

$$3^3 = 27 \Rightarrow \log_3 27 = 3$$

EXAMPLE B

Domain and asymptote
of $h(x) = \log_3(x + 4)$

$$x + 4 > 0 \Rightarrow x > -4$$

Domain $(-4, \infty)$, asymptote $x = -4$

EXAMPLE C

End behavior: as
 $x \rightarrow 0^+$, $\log_3 x \rightarrow ?$

$$\log_3 \frac{1}{3^n} = -n$$

as $x \rightarrow 0^+$, $\log_3 x \rightarrow -\infty$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Average rate of change of a logarithm

$$\text{AROC on } [a, b] = \frac{f(b) - f(a)}{b - a} \quad \log_3 x \text{ is increasing, but its rate of change falls}$$

EXAMPLE A

AROC of
 $f(x) = \log_3 x$ on $[1, 3]$

$$\frac{\log_3 3 - \log_3 1}{3 - 1} = \frac{1 - 0}{2}$$

$$= \frac{1}{2}$$

EXAMPLE B

AROC of
 $f(x) = \log_3 x$ on $[3, 9]$

$$\frac{\log_3 9 - \log_3 3}{9 - 3} = \frac{2 - 1}{6}$$

$$= \frac{1}{6}$$

EXAMPLE C

AROC of 3^x on $[0, 2]$
and of $\log_3 x$ on $[1, 9]$

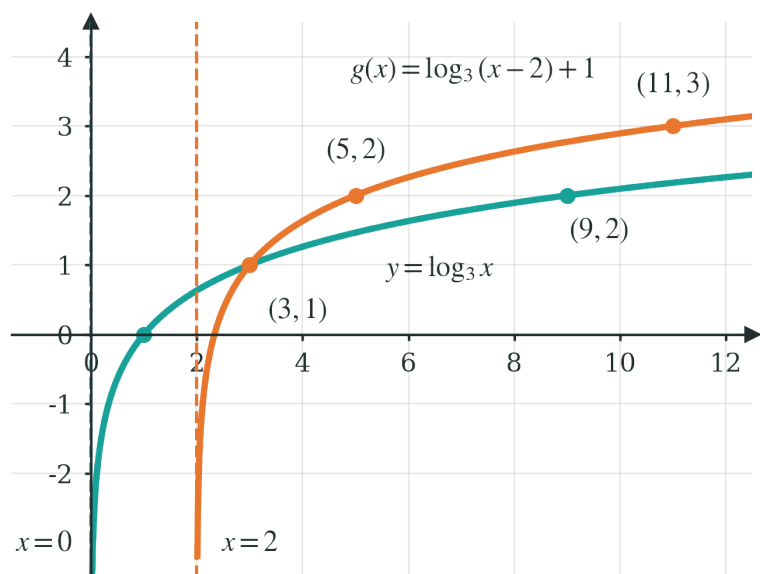
$$3^x : \frac{9 - 1}{2 - 0} = 4$$

$$\log_3 x : \frac{2 - 0}{9 - 1} = \frac{1}{4}$$

STATION 2 • TARGETED INSTRUCTION • 18 MIN

Graphing transformations of a logarithm

$$g(x) = \log_3(x - h) + k, \quad \text{asymptote } x = h, \quad \text{domain } x > h \quad \text{each point } (x, y) \rightarrow (x + h, y + k)$$



EXAMPLE A

Graph $g(x) = \log_3(x - 2) + 1$

Asymptote $x = 2$, domain $x > 2$

$(3, 1)$, $(5, 2)$, $(11, 3)$

EXAMPLE B

Write: $\log_3 x$
moved left 4, down 2

$$y = \log_3(x + 4) - 2$$

Asymptote $x = -4$; $(-3, -2)$ on graph

STATION 3 • PRACTICE • 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Evaluate $\log_2 32$ and $\log_5 \frac{1}{25}$

Find the inverse of $f(x) = 2^x - 3$

Done when: both logarithms are evaluated by writing the matching exponential, and the inverse is checked with one point.

APPLY

Use it in context

Average rate of change of $\log_3 x$ on $[3, 27]$

Domain and asymptote
of $h(x) = \log_3(x + 1) - 4$

Done when: the rate is written as a simplified fraction with its meaning in a sentence, and the domain uses interval notation.

INVESTIGATE

Find out why

Why can $\log_3 x$ never cross the line $x = 0$?

AROC of $\log_3 x$ on $[a, 3a]$: same for every a ?

Done when: you give a reason using the exponential with base 3, and a test with at least two values of a .

STATION 4 • PRODUCTION • 12 MIN

Production: the Red Sea tourism fund

THE MODEL

A model fund for a Red Sea tourism project triples in value every five years. Using model values only, its value in millions of SAR after t five-year periods is 3 to the power t , starting from 1 million SAR. Investors want to know how long it takes to reach a target value.

$$V(t) = 3^t \text{ (SAR millions)}$$

$$t = \log_3 V \text{ (inverse of } V)$$

1. Make a table of V and t for $V = 9, 27$ and 81 , and write the inverse model.
2. Find the average rate of change of t on V from 9 to 81 and say what it means for the fund.
3. Explain why the graph of t against V has a vertical asymptote at $V = 0$.

DONE WHEN

- The table reads t from V correctly for $2, 3$ and 4 periods.
- The average rate of change is $1/36$ period per million SAR, with units.
- The meaning says that each extra million SAR adds only a little time, and that the rate keeps falling.
- The asymptote explanation uses the fact that the fund value is always positive.

Hint for solving: to find t , ask which power of 3 gives V ; for the rate, use change in t divided by change in V .

STATION 5 • MASTERY GATE • 6 MIN

Mastery Gate: three questions to pass

1

Find the inverse of $f(x) = 3^x + 5$

2

Domain and asymptote
of $g(x) = \log_3(x + 6)$

3

AROC of $\log_3 x$ on $[3, 81]$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 • SMART PRODUCTION • 6 MIN

Smart Production: make one, then check it

CREATE

1. Write one inverse question: give an exponential function with a vertical shift and ask for its inverse.
2. Write a full solution on the back, with the domain of the inverse and its asymptote.
3. Swap with a partner and solve theirs.

Done when: your question has a unique answer, and your solution shows the swap, the solving step and one checking point.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

Write the inverse of $f(x) = \log_3(x - 1)$
State the domain and asymptote of f

Next lesson: 6-5 Properties of Logarithms.

Exam connections

SAT

If $f(x) = 2^x + 1$, what is $f^{-1}(9)$?

STEPS

1 $2^x + 1 = 9$

2 $2^x = 8$

3 $x = 3$

GAT

A: $\log_3 27$ B: $\log_2 16$

STEPS

1 $A = 3 \ (3^3 = 27)$

2 $B = 4 \ (2^4 = 16)$

3 B is greater

SAAT

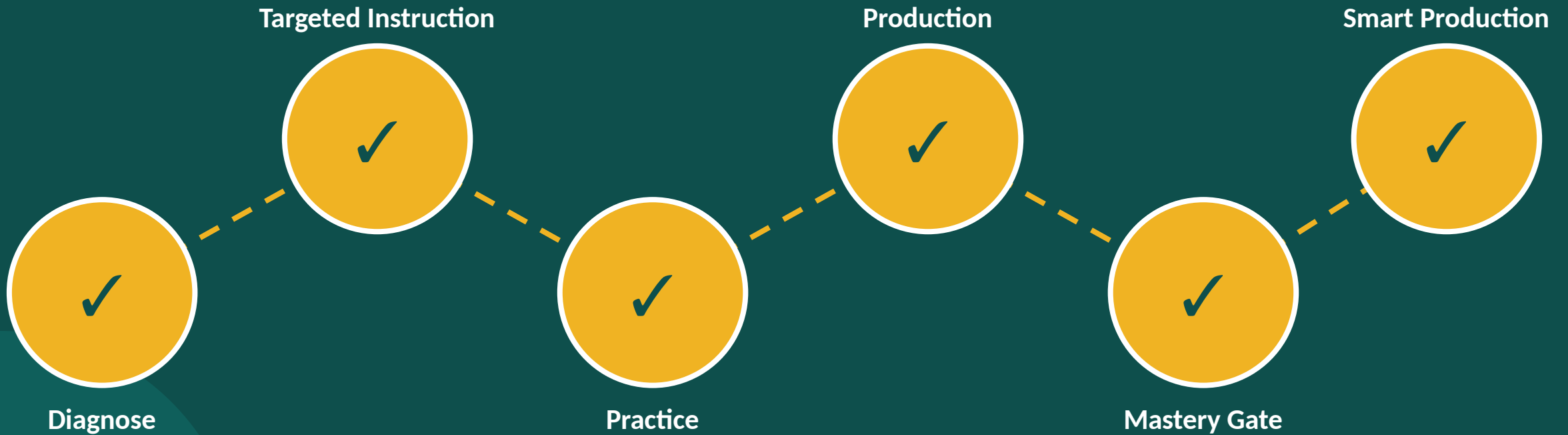
Vertical asymptote
of $y = \log_2(x + 5) - 7$

STEPS

1 $x + 5 = 0$

2 $x = -5$

Journey complete



A logarithm undoes an exponential, so its graph mirrors it across the line y equals x : domain x above 0, asymptote at x equals 0, and shifts move both.